FREEDOM INTERNATIONAL SCHOOL CHEMISTRY L-10 BIOMOLECULES GRADE XII	
1.	Write the structure of the product obtained when D-glucose is oxidised with nitric acid.
2.	Write chemical reactions to show that the open-chain structure of D - glucose contains the following: a) an unbranched carbon chain of six carbon atoms b) five alcohol groups c) an aldehyde group
3.	What are the expected products of hydrolysis of lactose ?
4.	What is a glycosidic linkage?
5.	State is the basic structural difference between glucose and fructose?
6.	Give the Haworth projection of a) D – glucopyranose b) D – fructofuranose c) Sucrose d) Maltose e) Lactose
7.	Write the name of linkage joining two amino acids.
8.	Name the deficiency diseases resulting from lack of Vitamins A and E in the diet.
9.	Name one fat soluble vitamin which is a powerful antioxidant and give its one natural source.
10.	Name a water soluble vitamin which is a powerful antioxidant. Give its one natural source.
11.	What are three types of RNA molecules which perform different functions?
12.	Which vitamin is synthesised in the human body in the presence of sunlight? Name the disease caused by its deficiency.
13.	Explain what is meant by (i) a peptide linkage (ii) a glycosidic linkage (iii) invert sugar (iv) pyranose structure of glucose (v) anomers (vi) polysaccharides (vii) essential amino acids
14.	What are enzymes?
15.	Name the four bases present in DNA. Which one of these is not present in RNA?
16.	Name the products of hydrolysis of sucrose. Why is sucrose not a reducing sugar?
17.	What are essential and non-essential amino acids in human food? Give one example of each type.
18.	Name the bases present in RNA. Which one of these is not present in DNA?
19.	Write the main structural difference between DNA and RNA.
20.	Mention the reactions and facts about glucose which cannot be explained by its open chain structure.
21.	A nucleotide from DNA containing thymine is hydrolysed. What are the products found?
22.	Explain the following: a) Amino acids behave like salts rather than simple amines or carboxylic acids. b) The two strands of DNA are complementary to each other. c) Starch and cellulose both contain glucose unit as monomer, yet they are structurally different.

23.	Differentiate between a) fibrous proteins and globular proteins b) amylose and amylopectin c) peptide linkage and glycosidic linkage d) nucleoside and nucleotide e) α-helix and β-pleated structures of proteins.
24.	What is meant by the denaturation of a protein?
25.	What is the other name of vitamin B_6?
26.	Name the vitamin whose deficiency causes increased blood clotting time.
27.	Draw a representative structural unit of a) Starch b) Nucleotide c) Nucleoside
28.	What is isoelectric point?
29.	A non-reducing disaccharide 'A' on hydrolysis with dilute acid gives an equimolar mixture of D-(+)-glucose and D-(-)-Fructose. Identify A. What is the mixture of D-(+)- glucose and D-(-)-Fructose known as? Name the linkage that holds the two units in the disaccharide.
30.	What is a zwitterion? Explain how it is formed.